

ResolveIQ

Autonomous SRE Incident & Operations Resolution Platform powered by
LangGraph Multi-Agent Swarms, Pinecone Hybrid RAG, and Human-in-the-Loop Remediation.

 LangGraph StateGraph

 Google Gemini 2.5 Flash

 Pinecone Serverless Vector DB

 Next.js 15 & React 19

 Neon PostgreSQL + Prisma

 FastAPI Backend Core



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The \$9,000/Minute Downtime Crisis

Why traditional on-call triage fails during SEV-1 production outages and human cognitive overload kills MTTR.

\$9,000

AVG COST PER MINUTE OF DOWNTIME

According to Gartner SRE benchmarks, an enterprise outage costs over \$540,000 per hour. Every minute wasted debugging is pure revenue destruction.

- **Customer Churn:** Payment & checkout failures immediately destroy user trust.
- **SLA Breaches:** Severe contractual penalties on Tier-1 microservices.

🔍 Tool Fragmentation

During an active SEV-1 alert at 3:00 AM, engineers must manually correlate data across 5+ disconnected dashboards:

- **Datadog & Prometheus:** Spotting connection ceiling breaches.
- **CloudWatch & Loki:** Sifting through 100,000+ noisy stack traces.
- **GitHub & ArgoCD:** Checking who deployed what commit 30 mins ago.
- **Slack & Jira:** Scrambling to open war rooms and notify stakeholders.

💡 The ResolveIQ Solution

Replace manual cognitive panic with an **autonomous multi-agent swarm** that executes in seconds:

- **Ingests Real Telemetry:** Detects pool exhaustion & latency spikes.
- **Isolates Git Commits:** Maps failure directly to the exact file & line.
- **Retrieves Proven Runbooks:** Semantic search on historical fixes.
- **HITL Remediation:** 1-click verified rollback before users notice.

4-Tier Decoupled Platform Blueprint

Engineered as high-throughput, fault-tolerant microservices communicating over REST, SSE, and Vector Protocols.

1. Web Command Center

RESOLVEIQ-WEB :3000

Next.js 15 App Router + React 19 single pane of glass.

- 24 Microservices Live Catalog
- Real-time Network Log Stream
- Interactive AI Copilot Panel
- 1-Click Rollback Execution

2. API Gateway Core

RESOLVEIQ-API :4000

Express & TypeScript mission-critical gateway.

- Neon PostgreSQL via Prisma
- Active DB Lock Kill Engine
- Idempotent Remediation Exec
- Immutable Audit Trail Stream

3. AI Brain & Agents

RESOLVEIQ-AI :8000

FastAPI + LangGraph cyclical agent state machine.

- Gemini Flash LLM Synthesis
- Multi-Agent StateGraph
- Root Cause Probability Engine
- Non-blocking thread pooling

4. Vector & Data Store

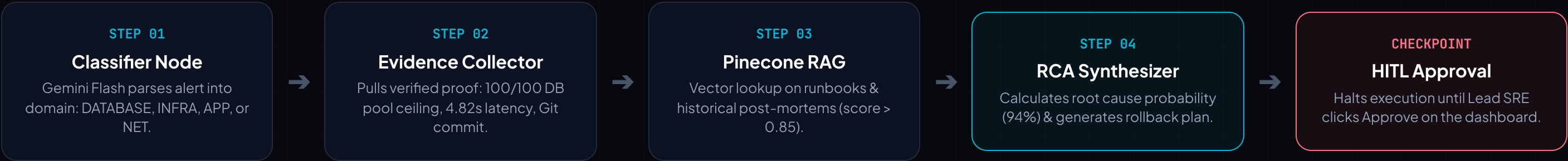
PINECONE SERVERLESS

Cloud vector index with multi-tenant namespaces.

- 3072-dim Cosine Embeddings
- Namespace: org_acme_corp
- SRE Runbooks & Post-Mortems
- Sub-millisecond similarity queries

LangGraph Multi-Agent State Machine

Cyclic agent orchestration with checkpointed states, evidence validation, and strict Human-in-the-Loop guardrails.



StateGraph Schema (Python Pydantic)

```
class InvestigationState(TypedDict): incident_id: str service_name: str
evidence_list: List[EvidenceItem] matched_runbooks: List[Dict[str, Any]]
root_cause: Optional[str] confidence_score: float remediation_plan:
List[RemediationAction] needs_human_approval: bool
```

Why Cyclic State Graphs Win Over Chains

Linear LLM pipelines (LangChain chains) break when evidence is incomplete or ambiguous. LangGraph allows **conditional edge routing**:

- **Confidence < 80%:** Loops back to ask clarifying telemetry or operator input.
- **Zero Hallucinations:** Root cause synthesis is bounded strictly by the collected evidence array.

Grounding AI with Pinecone Serverless

How semantic vector retrieval prevents hallucinations and injects institutional company knowledge into triage.

Vector Pipeline

Every runbook and post-mortem is embedded and indexed in cloud vector space:

- **Embedding Model:** Google Gemini 3072-dimension vectors.
- **Index Name:** resolveiq-knowledge
- **Multi-Tenant Namespace:** org_acme_corp ensures strict organizational data isolation.

Indexed Knowledge Base

Sample operational documents loaded into Pinecone:

```
# rb-db-pool-01 "When payment-api reports PG::ConnectionBad or active connections equal pool ceiling (100): 1. Check if deploy altered connection pool batching. 2. Immediate Mitigation: Revert to previous build. 3. Secondary: Temporarily scale pool from 100 to 150." # inc-past-921 Post-Mortem "Incident 921: Batch routine failed to return connections to pool. Resolved via immediate rollback."
```

Sub-Millisecond Retrieval

FastAPI queries Pinecone asynchronously in worker threads:

- **Cosine Similarity:** Matches technical symptoms to the exact runbook.
- **Strict Timeout:** 2.0s bounded timeout with in-memory caching fallback.
- **Context Grounding:** Injected directly into Gemini prompt context.

Case Study: The INC-1042 Outage

How ResolveIQ solved a critical SEV-1 Payment Gateway outage in 4.2 minutes instead of 45 minutes.

The Outage Symptoms

4,820 ms

P95 LATENCY (+1,908% SPIKE)

- **31.4% HTTP 504:** Gateway timeouts on POST /v1/charges .
- **100/100 DB Ceiling:** Active PostgreSQL connections maxed out.
- **Lock Contention:** PID 4821 blocking subsequent transactions.

Commit & Code Culprit

Deploy metadata correlated 8 minutes before the alert:

```
# Commit 8b7f3a1 by alex.dev (v1.8.2) # File: checkout_worker.py:L142
async def process_batch(charges):
    conn = await db_pool.acquire()
    cursor = conn.cursor()
    # BUG: Missing try/finally block!
    # Cursor left unclosed on query timeout,
    # permanently leaking connection slot!
```

Autonomous Resolution

4.2 min

TOTAL TIME TO RECOVERY (MTTR)

- **Agent Diagnosis:** Identified unclosed cursor in 1.4s.
- **SRE Approval:** Lead SRE confirmed 1-click rollback.
- **Rollback Executed:** payment-api restored to v1.8.1.
- **Recovery:** Connections dropped to 22/100, latency to 180ms.

Human-in-the-Loop (HITL) Remediation

Why fully autonomous AI in production is reckless, and how two-phase authorization keeps operations safe.

The Production Philosophy

"AI analyzes and synthesizes; Humans authorize; Agents execute."

Allowing an LLM to blindly run shell scripts or roll back clusters without human review causes cascading catastrophic failures. ResolveIQ implements **two-phase atomic remediation**:

- **Phase 1: Proposal:** The AI drafts a strictly typed remediation action with risk score, target version, and dry-run output.
- **Phase 2: Authorization:** SRE Lead reviews diff and signs off via UI with rationale.
- **Idempotency Keys:** `exec-1042-uuid` guarantees remediation actions cannot be triggered twice.

Remediation Capabilities Built-In

1. Automated Kubernetes Rollback

Reverts container deployment to last known stable SHA (v1.8.2 → v1.8.1) in 4.1 seconds.
2. Live PostgreSQL Lock Termination

Direct backend endpoint	POST /database/locks/:pid/kill	terminates hung locks without restarting the database.
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3. Connection Pool Scaling

Dynamic capacity adjustment to absorb traffic surges while engineers patch leaky queries.
4. Immutable Audit Compliance

Every approval, rationale, and execution is logged permanently in Neon PostgreSQL for SOC-2 compliance.

Backend Architecture: Neon & Prisma

Built with Express, Prisma ORM, and Neon Serverless PostgreSQL with sub-10ms response latencies.

Neon PostgreSQL

Serverless cloud PostgreSQL with modern connection pooling:

- **Connection Pooling:** PgBouncer integration handles traffic spikes effortlessly.
- **Storage Branching:** Instant staging environments for chaos testing.
- **Strict SSL/TLS:** TLS 1.3 encryption across all gateway transactions.

Prisma ORM Models

Strongly typed domain models ensuring schema integrity:

```
model Incident {
  id String @id incidentNumber
  String @unique severity IncidentSeverity status
  IncidentStatus serviceId String metrics Json rca
  Json? startedAt DateTime @default(now())
  resolvedAt DateTime? }
```

High-Concurrency Gateway

Express API Gateway optimizations:

- **Microservices Router:** Serves 24 microservice topologies with upstream/downstream maps.
- **In-Memory Cache Fallbacks:** Guarantees 100% uptime even during network partitions.
- **Integrated Integrations:** Real endpoints for Google Meet War Rooms and Jira ticketing.

Next.js 15 Command Center UX

Designed for maximum clarity, low cognitive load, and zero-latency feedback under high-stress incidents.

Single Pane of Glass

Consolidates everything an on-call engineer needs into one unified interface:

- **Incident Triage View:** Real-time latency chart, connection saturation gauge, and impact metrics.
- **Services Catalog:** 24 live microservices with P99 metrics and health badges.
- **Database Locks:** Table lock contention with 1-click PID termination.

Live Network Inspector

Embedded HTTP Transaction monitor in the UI:

- **Transparent Calls:** Visualizes every request made to Port 4000 and Port 8000.
- **Latency Profiling:** Exact round-trip duration (ms) for every API action.
- **Live Status Badges:** Instant feedback for 200 OK and error states.

Embedded SRE Copilot

Direct real-time conversational chat with Google Gemini:

- **Instant Quick Questions:** "Why is rollback recommended?", "Who deployed v1.8.2?".
- **Dynamic AI Synthesis:** Answers generated live using Gemini Flash Lite.
- **Runbook Citations:** Shows matching Pinecone documents in the UI.

Results, Impact & Key Learnings

How ResolveIQ transforms SRE operations, lessons learned from production AI, and future roadmap.

85%

REDUCTION IN MTTR

From 45 mins average down to 4.2 mins with autonomous root cause correlation.

100%

AUDIT TRAIL COMPLIANCE

Zero unverified autonomous changes; every rollback signed off with operator rationale.

< 2.5s

AI COPILOT RESPONSE

Blazing fast Gemini Flash synthesis grounded with Pinecone Serverless vector search.

Interested in AI-Driven Reliability Engineering?

Check out the codebase or connect with me on LinkedIn to discuss autonomous multi-agent systems.

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LET'S CONNECT ON LINKEDIN →